Explanation of Files

1. total_sarimax.joblib

What it is: A serialized Python object containing the trained SARIMAX model used to forecast the total monthly tea production across all elevations and tea types.

SARIMAX (Seasonal Autoregressive Integrated Moving Average with eXogenous regressors): A statistical time-series forecasting model that captures trends, seasonality, and also incorporates external variables such as Rainfall, Humidity, and Air Temperature.

Why it is saved: The model is stored using joblib so it can be loaded quickly without retraining. This is essential for deployment environments such as production servers and is responsible for predicting the overall market size.

2. share_catboost.joblib

What it is: A serialized Python object containing the trained CatBoostRegressor model used to predict the monthly share of total production for each specific combination (for example: HIGH – GREEN TEA).

CatBoostRegressor: A gradient boosting machine learning algorithm that performs well with categorical variables and typically provides high predictive accuracy.

Why it is saved: This file allows the trained model to be loaded directly for prediction. The model distributes the total production forecast (generated by the SARIMAX model) across different tea combinations based on their predicted shares.

3. metadata.json

What it is: A JSON file that stores structured and human-readable information about the entire World Model, particularly the version that incorporates structural zeros.

Contents include: Project details such as name and export date; data definitions including target variables, combination definitions, lists of exogenous variables, and the set of all combinations used by the model. It also records structural-zero combinations that are expected to always have zero production.

Model parameters: Information about the models such as the SARIMAX configuration (order, seasonal order, seasonal period, and exogenous variables) and details of the CatBoost share model including feature columns and categorical variables.

Feature engineering: Descriptions of how model features were created, including lag variables, rolling windows, monthly encoding, share targets, and normalization logic.

Reproducibility: Includes random seeds and other parameters that allow the forecasting system to reproduce the same results consistently.

Why it is saved: The metadata file acts as documentation and configuration for the forecasting system. It helps with understanding the model structure, ensuring correct deployment, and maintaining reproducibility of results.

Summary

The .joblib files represent the executable components of the forecasting system. The total_sarimax.joblib model predicts the overall tea production, while the share_catboost.joblib model distributes that production among specific tea categories. The metadata.json file serves as documentation and configuration for the system, ensuring correct usage, reproducibility, and easier deployment.